

ALY 6040 Data Mining Applications

Module 2 - Discussion

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Proposed Dataset Description

The decision tree provides a detailed classification model predicting whether borrowers from LendingClub.com will repay their loans or default.

a. Feature Importance Analysis

From the decision tree, the top three most important features seem to be:

- i. **int.rate (Interest Rate):** The interest rate represents the cost of borrowing for the applicant. Higher interest rates are often associated with riskier borrowers, making this an essential factor in assessing default risk.
- ii. **fico (FICO Score):** A higher FICO score is indicative of a stronger credit history, thus borrowers with lower FICO scores are more likely to default. This is consistently one of the first splits in the tree, signifying its importance.
- iii. **dti (Debt-to-Income Ratio):** This measures the borrower's debt as a percentage of their income. A high DTI indicates financial strain, as a larger proportion of income is allocated to debt repayment, increasing the likelihood of default.

These features are critical as they directly relate to the borrower's ability to manage debt and reflect their creditworthiness.

b. Decision Path Interpretation

Path leading to loan default:

- i. Starting from the root: $\text{credit.policy} \leq 0.5$, meaning borrowers who don't meet the company's credit policy are examined further.
- ii. The next node checks if $\text{int.rate} \leq 0.093$. If true, the borrower might still be considered low risk.
- iii. Further down, $\text{fico} \leq 741.0$, meaning that a lower credit score is a signal of higher risk.
- iv. Finally, the node $\text{int.rate} \leq 0.08$ results in a high likelihood of default, as low FICO score combined with certain interest rates make repayment less likely.

Each split refines the borrower's profile, and the final path's conditions (low FICO score and high interest rate) strongly indicate default risk.

c. Threshold Impact Evaluation

Take the node with $\text{fico} \leq 657.5$ as an example. Changing this threshold to a higher value (e.g., 670) would include more borrowers in the high-risk category. This adjustment could increase the model's sensitivity (identifying more defaults) but might reduce specificity (correctly predicting borrowers who will repay). Such a change could lead to more false positives, where borrowers who would have repaid are now misclassified as likely defaulters, thus affecting the model's precision.

d. Model Limitations Discussion

Several limitations may be present in this decision tree:

- i. Overfitting: The tree may be too specific to the training data, especially given the multiple splits with small sample sizes at the leaf nodes. This could result in the model performing poorly on unseen data.
- ii. Bias in FICO Scores: The reliance on FICO scores and interest rates might lead the model to unfairly penalize groups of borrowers who have lower FICO scores due to systemic issues rather than actual credit risk.
- iii. Granularity in splits: Some splits, such as precise thresholds for `intr.rate`, might not generalize well outside the dataset and may need to be adjusted dynamically based on market conditions.

e. Predictive Accuracy Improvement

Two suggestions for improving the model's predictive accuracy:

- i. Feature Engineering: Create additional features like `credit.age` (age of credit history) or interaction terms between `dti` and `fico`. These might capture borrower profiles better than individual features alone.
- ii. Pruning: Pruning the tree to prevent overfitting can lead to better generalization. Alternatively, using techniques like cross-validation during tree construction to stop growth at an optimal point might improve performance on new data.

References:

- a. Lending Club. (n.d.). Loan Data. Retrieved from <https://www.lendingclub.com/info/download-data.action>
- b. Scikit-learn Developers. (2023). Decision Trees. In Scikit-learn 1.3.0 documentation. <https://scikit-learn.org/stable/modules/tree.html>